

Modern coexistence theory

Learning objectives

1. Know the terminology used in MCT and how it relates to processes we've already discussed

last lecture: processes that lead to coexistence and diverse communities, general

today, specifically talking about modern coexistence theory

Modern coexistence theory

- coexistence: fitness differences, large niche differences
- niche differences: species differences that make species limit itself more than others *processes we talked about last week*
- fitness differences: species differences that determine outcome without niche differences *superior vs. inferior competitor*

how do you keep niche differences large and fitness differences small?

Mechanisms

- storage effect - environmental fluctuations, store seeds in good years, other plants grow in bad years

this assumes that species have different good years and bad years, ties back to niche differentiation, but more about storing success

- relative nonlinearity - relationship between fitness and resource level not all same *in other words, not all linear, they have different shapes*
- fitness-density covariance - intraspecific aggregation *fitness highest where other conspecifics are, requires spatial heterogeneity*

Pros and cons

- advantages: quantitative, strength of mechanisms
- disadvantages: fitness/niche differences easier than mechanisms